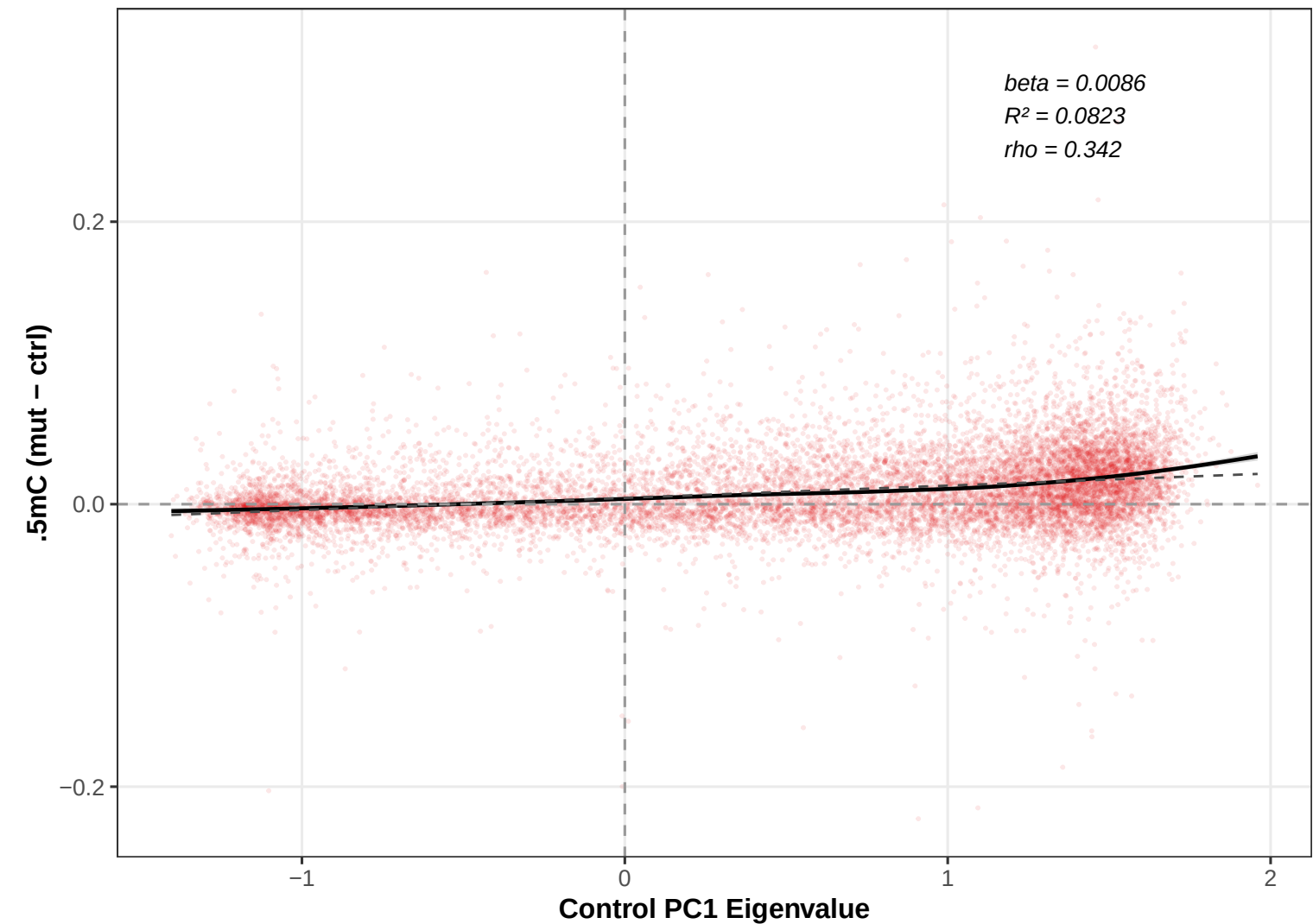


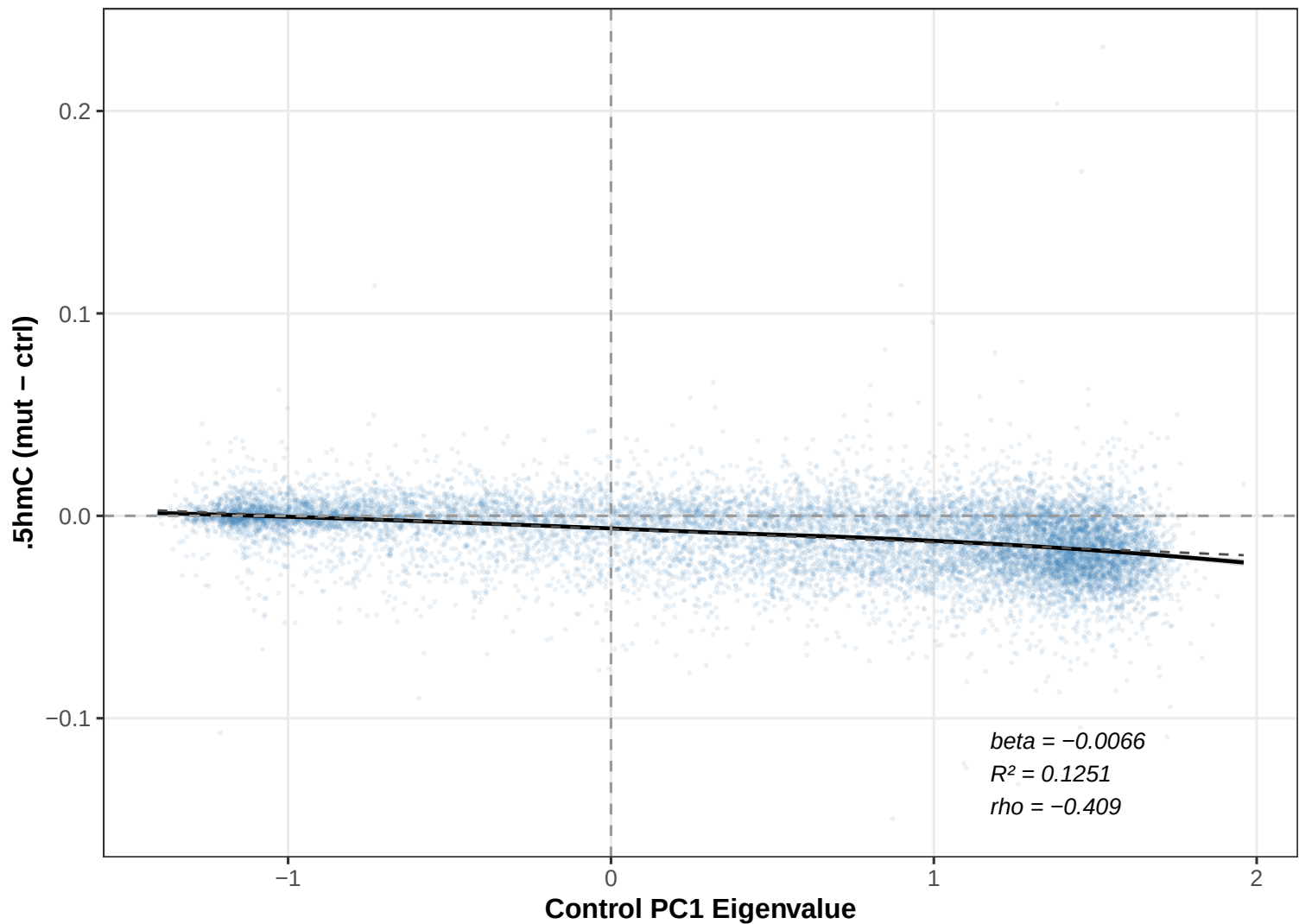
Mechanism Differentiation: TET Impediment vs DNMT3A Recruitment

Compartment-dependent gradients, mediation, and variance partition

5mC Change vs Control PC1

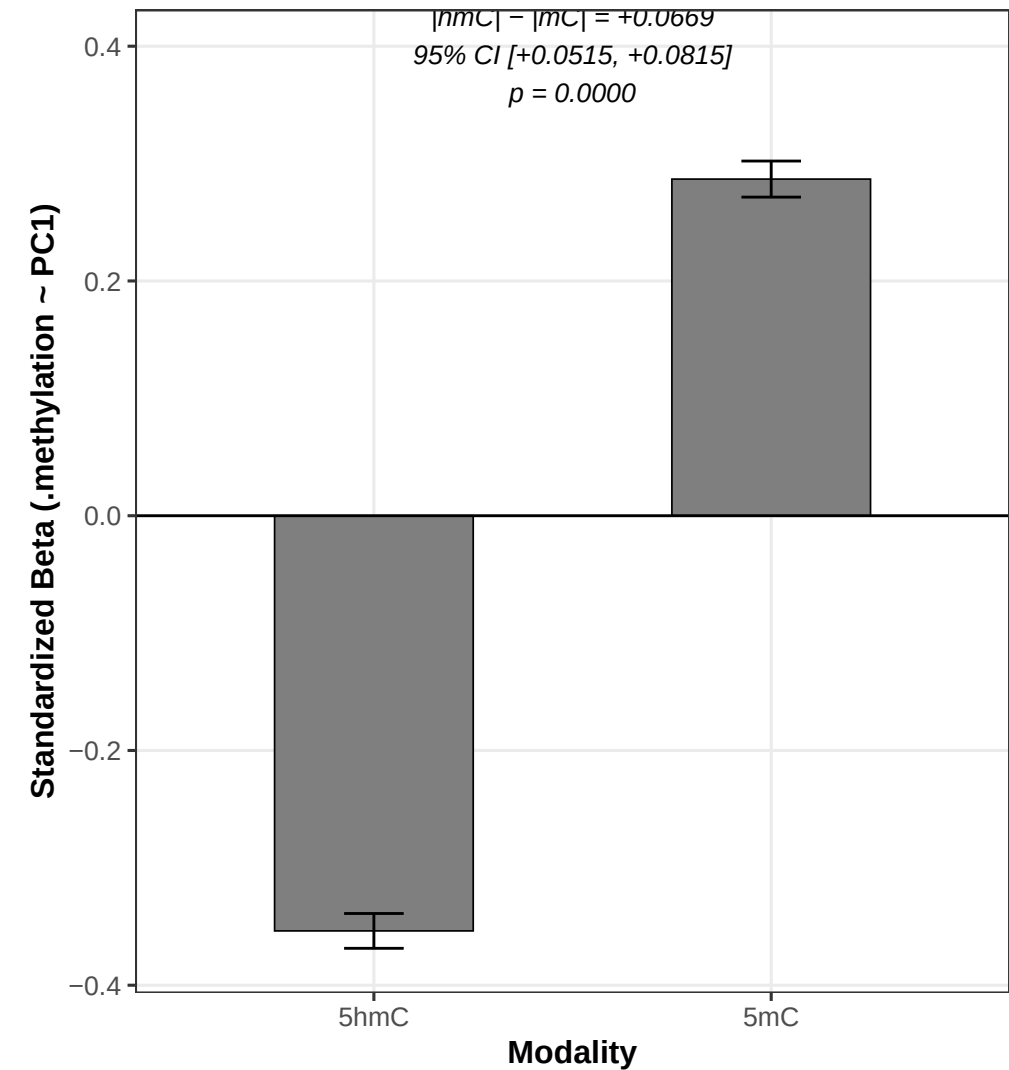


5hmC Change vs Control PC1



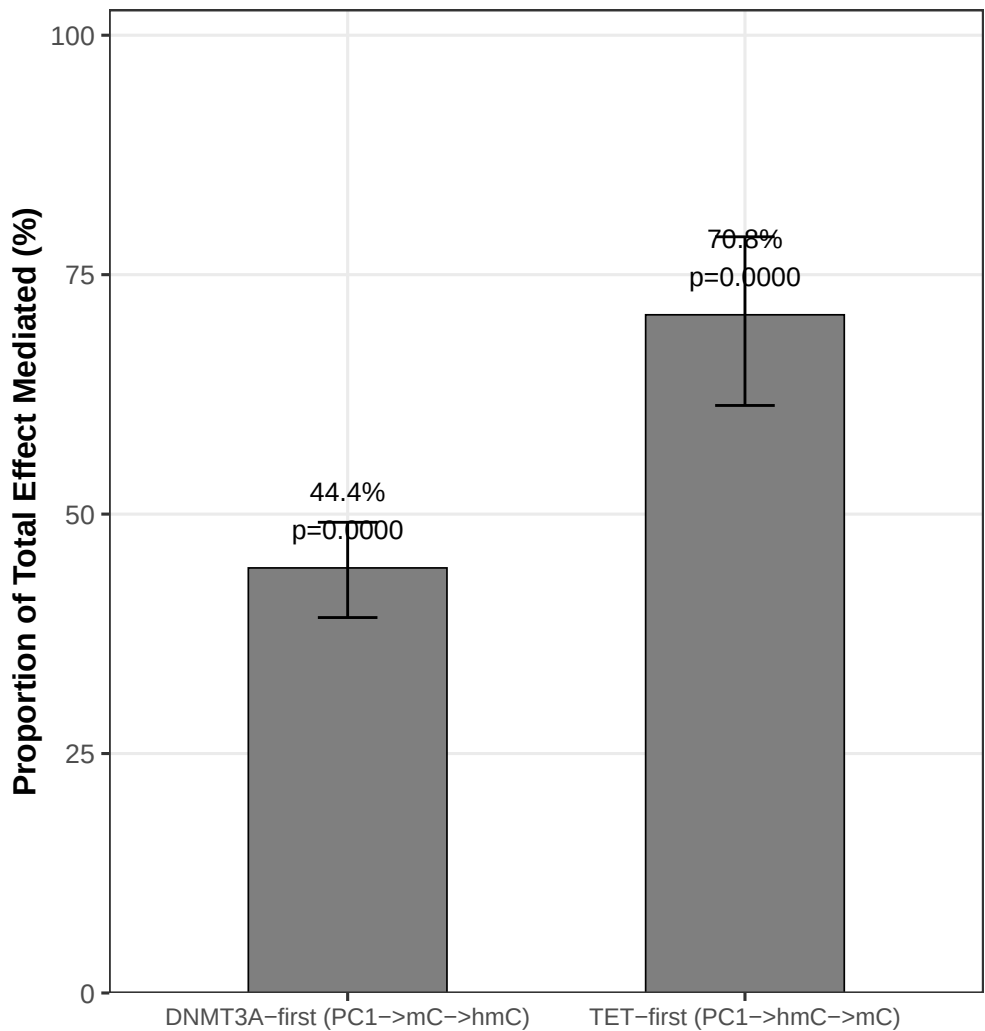
Compartment Gradient Steepness: 5mC vs 5hmC

Standardized beta from PC1 regression (bootstrap 95% CI)



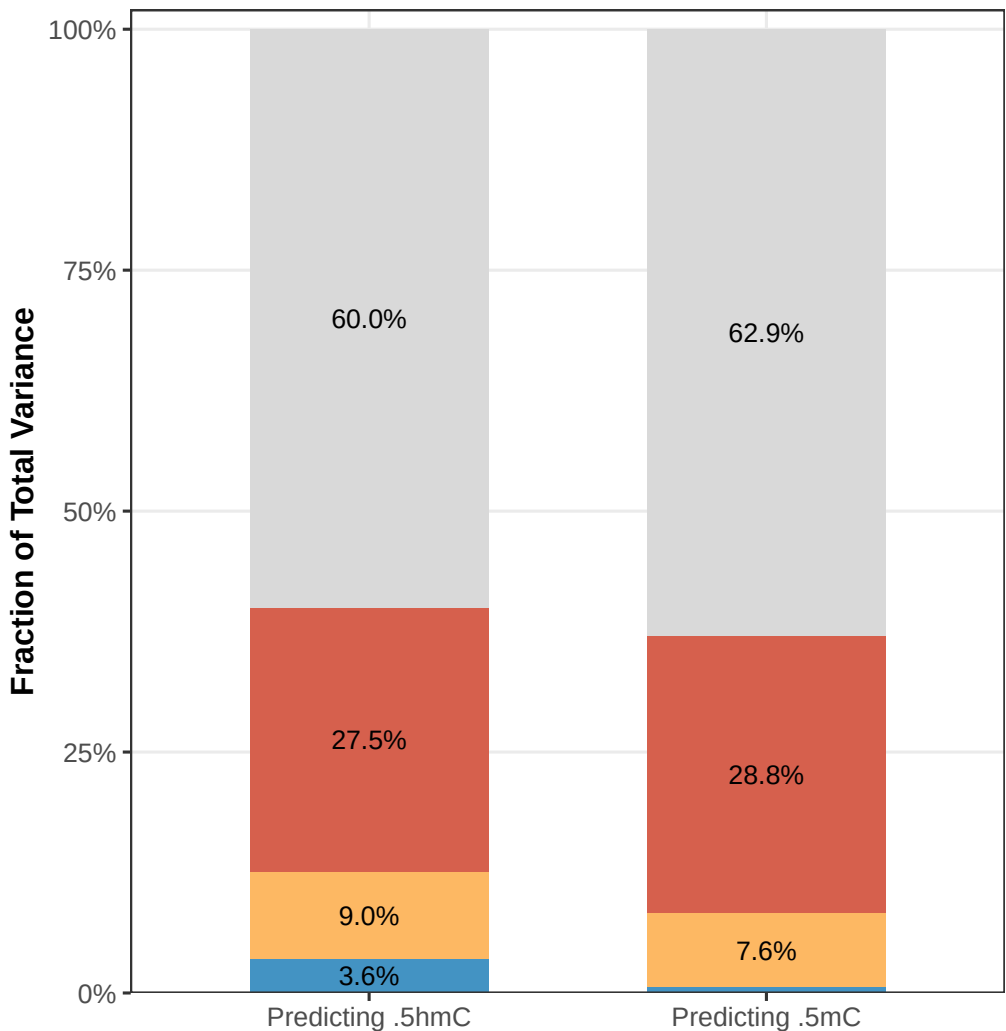
Mediation Analysis: Which Mechanism Mediates the PC1 -> Methylation Link?

Higher proportion mediated = stronger evidence for that causal path



Variance Partition: PC1 vs Other Modality

How much variance does compartment identity explain uniquely?



Component

- Unexplained
- Other modality unique
- Shared
- PC1 unique

Methylation Changes in B-Compartment Genes

B-cmpt Fisher OR (mC sig x hmC sig) = 6.64, $p < 2.2e-16$

